Mathematics for New Technologies in Finance : Project

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PART I

Question 1

The set Θ_0 consists of all trading strategies $(\theta_t)_{t\geq 0}$ given by $\theta_t = \nabla\varphi(X_t)$, where X_t is the traded asset on $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ and where φ belongs to the class of functions $\mathcal{D}$. In particular, $\varphi : E \subset \mathbb{R}^d \to \mathbb{R}$ must be twice continuously differentiable on E and must satisfy:

$$\int_E \left| \frac{\mathrm{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} \right| dx < \infty$$

where $A : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ is given by $A(x) = cp(x)$. Here, $c \in \mathbb{S}^d$ is a symmetric PD matrix that defines the convariance of the Ornstein-Uhlenbeck process X_t and $p(x)$ denotes the density of the d-variate normal distribution centered at 0 with correlation matrix Σ

Question 2

We first rewrite $\log V_T^{\theta_0}$:

$$\log V_T^{\theta_0} = \log \mathcal{E}\left(\int_0^T \theta_t dX_t \right) \tag{1}$$

$$= \int_0^T \theta_t dX_t - \frac{1}{2}\left\langle \int_0^T \theta_t dX_t \right\rangle \tag{2}$$

$$= \int_0^T \nabla\varphi(X_t) dX_t - \frac{1}{2}\left[\int_0^{\cdot} \nabla\varphi(X_t) dX_t \right]_T \tag{3}$$

where in the last line we have used that $\nabla\varphi$ is continuous, and so are the trajectories of the OU process X_t, resulting in the optional variation $\langle\cdot\rangle$ to turn into the quadratic variation $[\cdot]$. We now apply Itô's lemma to obtain an expression for $d\varphi(X_t)$:

$$d\varphi(X_t) = \sum_{i=1}^d \partial_{x_i}\varphi(X_t) dX_t^i + \frac{1}{2}\sum_{i=1}^d \sum_{j=1}^d \partial_{x_i}\partial_{x_j}\varphi(X_t) d[X^i, X^j]_t \tag{4}$$

$$= \nabla\varphi(X_t) dX_t + \frac{1}{2}\mathrm{Tr}(\nabla^2\varphi(X_t) d[X]_t) \tag{5}$$

Without detailing the computations of the quadratic variation of the integral term in (3), and plugging (5) inside (3), we obtain

$$\log V_T^{\theta_0} = \int_0^T d\varphi(X_t) - \frac{1}{2}\int_0^T \sum_{i=1}^d \sum_{j=1}^d \partial_{x_i}\partial_{x_j}\varphi(X_t) d[X^i, X^j]_t - \frac{1}{2}\int_0^T \sum_{i=1}^d \sum_{j=1}^d \partial_{x_i}\varphi(X_t)\partial_{x_j}\varphi(X_t) d[X^i, X^j]_t \tag{6}$$

$$= \int_0^T d\varphi(X_t) - \frac{1}{2}\int_0^T \sum_{i=1}^d \sum_{j=1}^d (\partial_{x_i}\partial_{x_j}\varphi(X_t) + \partial_{x_i}\varphi(X_t)\partial_{x_j}\varphi(X_t)) d[X^i, X^j]_t \tag{7}$$

Some manipulation of the double sums in the intagrand of (7), as well as an omitted computation (see appendix) of $d[X^i, X^j]_t = c_{ij}dt$ yield the following:

$$\log V_T^{\theta_0} = \int_0^T d\varphi(X_t) - \frac{1}{2}\int_0^T \mathrm{Tr}[c(\nabla^2\varphi(X_t) + \nabla\varphi(X_t)\nabla\varphi(X_t)^\intercal)]dt \tag{8}$$

$$= \varphi(X_T) - \varphi(X_0) - \frac{1}{2}\int_0^T \mathrm{Tr}[c(\nabla^2\varphi(X_t) + \nabla\varphi(X_t)\nabla\varphi(X_t)^\intercal)]dt \tag{9}$$

Equation 9 can be further simplified by noting that the matrix $\nabla^2\varphi(X_t) - \nabla\varphi(X_t)\nabla\varphi(X_t)^\intercal$ can be rewritten using the following trick, introducing the Hessian of $e^{\varphi(x)}$:

$$[\nabla^2 e^{\varphi(x)}]_{ij} = \partial_{x_i}\partial_{x_j} e^{\varphi(x)} = \partial_{x_i}(\partial_{x_j}\varphi(x)e^{\varphi(x)}) = \partial_{x_i}\partial_{x_j}\varphi(x)e^{\varphi(x)} + \partial_{x_i}\varphi(x)e^{\varphi(x)}\partial_{x_j}\varphi(x) \tag{10}$$

$$= e^{\varphi(x)}[\partial_{x_i}\partial_{x_j}\varphi(x) + \partial_{x_i}\varphi(x)\partial_{x_j}\varphi(x)] \tag{11}$$

$$= e^{\varphi(x)}[\nabla^2\varphi(X_t) + \nabla\varphi(x)\nabla\varphi(x)^\intercal]_{ij} \tag{12}$$

Therefore, we have the simplification:

$$\frac{\nabla^2 e^{\varphi(x)}}{e^{\varphi(x)}} = \nabla^2\varphi(x) + \nabla\varphi(x)\nabla\varphi(x)^\intercal$$

Thus, we can rewrite 9 and divide by T to obtain:

$$\frac{1}{T}\log V_T^{\theta_0} = \frac{\varphi(X_T) - \varphi(X_0)}{2} - \frac{1}{2T}\int_0^T \frac{\mathrm{Tr}(c\nabla^2 e^{\varphi(X_t)})}{e^{\varphi(X_t)}}dt$$

To look at the asymptotic growth rate, we let $T \uparrow \infty$. Under the Ornstein-Uhlenbeck dynamics, the process X_t has ergodic (invariant) distribution $X_\infty \sim \mathcal{N}(0, \Sigma)$, for a given matrix Σ. The ergodic property gives:

$$\frac{1}{T}\int_0^T h(X_t)dt \xrightarrow{T\to\infty} \int_E h(x)p(x)dx \tag{13}$$

This property should be understood as "one very long trajectory of X_t will take on average the expected value of its ergodic density". We can proceed to rewrite our expression for the log-utility as:

$$\lim_{T\to\infty}\frac{1}{T}\log V_T^{\theta_0} = \lim_{T\to\infty}\frac{-1}{2}\frac{1}{T}\int_0^T \frac{\mathrm{Tr}(c\nabla^2 e^{\varphi(X_t)})}{e^{\varphi(X_t)}}dt \to \int_E \frac{\mathrm{Tr}(c\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}}p(x)dx \tag{14}$$

In the expression above, we can bring out the density $p(x)$ inside the trace, and letting $A \coloneqq cp(x)$, we have

$$\lim_{T\to\infty}\frac{1}{T}\log V_T^{\theta_0} = \frac{-1}{2}\int_E \frac{\mathrm{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}}dx$$

Question 3

We integrate by parts the expression:

$$\int_E \frac{\mathrm{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}}dx = \int_E \sum_{i=1}^d\sum_{j=1}^d \frac{A_{ij}}{u(x)}(\nabla^2 u)_{ji}dx$$

where we have rewritten $u(x) \coloneqq e^{\varphi(x)}$. Using the symmetry of the Hessian, and integrating by part for fixed indices (i, j), we have:

$$\int_{\partial E} A_{ij}\partial_i u\frac{1}{u}n_j(x)dS - \int_E \partial_j\left(\frac{A_{ij}}{u}\right)\partial_i u dx$$

Note that $\partial_i u = \underbrace{\partial_i\varphi(x)}_{=0 \text{ on } \partial E}\, e^{\varphi(x)}$ by smooth compactedness. Hence the first integral is 0 and for the second integral, we have:

$$-\int_E \partial_j(\frac{A_{ij}}{u})\partial_i u dx = -\int_E \left(\frac{\partial_j A_{ij}}{u} - \frac{A_{ij}\partial_j u}{u^2}\right)\partial_i u dx = -\int_E \partial_j A_{ij}\partial_i\varphi - A_{ij}\partial_i\varphi\partial_j\varphi dx =: I$$

Coming back to the sum over all indices i, j will yield

$$I = -\int_E \sum_{i=1}^d\sum_{j=1}^d \partial_j A_{ij}\partial_i\varphi dx + \int_E \sum_{i=1}^d\sum_{j=1}^d A_{ij}\partial_i\varphi\partial_j\varphi dx \tag{15}$$

$$= -\int_E (\mathrm{div}A)^\intercal\nabla\varphi dx + \int_E \mathrm{Tr}(A\nabla\varphi\nabla\varphi^\intercal)dx \tag{16}$$

$$= -\int_E (\mathrm{div}A)^\intercal\nabla\varphi dx + \int_E \mathrm{Tr}(\nabla\varphi^\intercal A\nabla\varphi)dx \tag{17}$$

$$= -\int_E (\mathrm{div}A)^\intercal\nabla\varphi dx + \int_E \nabla\varphi^\intercal A\nabla\varphi dx \tag{18}$$

where in the second-to-last equality we have used that the Trace operator is invariant under cyclic permutations (i.e. $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB) = \mathrm{Tr}(BCA)$)

Question 4

The divergence of a matrix field $A : \mathbb{R}^{d\times d} \to \mathbb{R}$ gives a vector field. In particular, we have have $(\mathrm{Div}A)_i = \sum_{j=1}^{d} \partial_j A_{ij}$. In particular:

$$\begin{aligned}(\mathrm{Div}A)_i &= \sum_{j=1}^{d} \partial_j A_{ij} \\ &= \sum_{j=1}^{d} \partial_j p(x) c_{ij} \\ &= [c]_i \cdot \nabla p \\ &\implies \mathrm{Div}A = c\nabla p\end{aligned}$$

where $[c]_i$ denotes the i-th row vector of the matrix c. Left-multiplying the last equation by the inverse of A (which exists since c is positive-definite and $p(x) > 0$) gives:

$$\begin{aligned}A^{-1}\mathrm{div}A &= A^{-1}c\nabla p \\ &= \frac{1}{p}\nabla p \\ &= \nabla \log p\end{aligned}$$

Question 5

From the previous question, we concluded that $\mathrm{Div}A = A\nabla \log p$. We plug this result in (18) to rewrite the integral as:

$$-\int_E (-A\nabla \log p)^\intercal \nabla\varphi dx + \int_E \nabla\varphi^\intercal A \nabla\varphi dx$$

Notice that:

$$\nabla \log p = \frac{-1}{2}(\Sigma^{-1}x + (\Sigma^{-1})^\intercal x) = -\Sigma^{-1}x$$

where we have used the facts that inverse and transpose operations are permutable, and that Σ is symmetric. which gives:

$$-\int_E (A\nabla \log p)^\intercal \nabla\varphi dx + \int_E \nabla\varphi^\intercal A \nabla\varphi dx = \int_E (x^\intercal \Sigma^{-1} + \nabla\varphi^\intercal) A \nabla\varphi dx$$

To summarize, the full robust optimization problem was to compute that asymptotic growth-rate $\sup_{\theta\in\Theta} \inf_{\mathbb{P}\in\Pi} g(\theta;\mathbb{P})$. We restricted our problem to the class of trading strategies $\Theta_0 = \{\theta_t : \theta_t = \nabla\varphi, \varphi \in \mathcal{D}\} \subset \Theta$, under which we have shown that $g(\theta;\mathbb{P})$ was independent of the measure $\mathbb{P}$. The optimization problem then reduced to : $\lambda^\star := \sup_{\varphi\in\mathcal{D}} g(\varphi) = \frac{-1}{2}\int_E \frac{\mathrm{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx$. Using our results of questions 4 and 5, we have now that:

$$\sup_{\varphi} \int_E \frac{\mathrm{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx = \sup_{\varphi\in\mathcal{D}} \int_E (x^\intercal \Sigma^{-1} + \nabla\varphi^\intercal) A \nabla\varphi dx$$

We now bring back the omitted so far factor of $\frac{-1}{2}$ and complete the square in the integral to obtain :

$$\begin{aligned}\lambda^\star &= \sup_{\varphi\in\mathcal{D}} \frac{-1}{2} \int_E (x^\intercal \Sigma^{-1} + \nabla\varphi^\intercal) A \nabla\varphi \, dx \\ &= \inf_{\varphi\in\mathcal{D}} \frac{1}{2} \int_E (x^\intercal \Sigma^{-1} + \nabla\varphi^\intercal) A \nabla\varphi \, dx \\ &= \inf_{\varphi\in\mathcal{D}} \frac{1}{2} \int_E \left[\left(\nabla\varphi + \frac{1}{2}\Sigma^{-1}x\right)^\intercal A \left(\nabla\varphi + \frac{1}{2}\Sigma^{-1}x\right) - \frac{1}{4}(\Sigma^{-1}x)^\intercal A (\Sigma^{-1}x)\right] dx\end{aligned}$$

Since A is a positive definite matrix and because the last term does not depend on the function φ, the minimizer of this optimization problem is naturally given by:

$$\nabla\varphi^\star(x) = \frac{-1}{2}\Sigma^{-1}x$$

Question 6

We conclude that the robust-optimal trading strategy is given by

$$\theta_t = \nabla\varphi(X_t) = \frac{-1}{2}\Sigma^{-1}X_t$$

Question 7

The robust-optimal growth rate is obtained by evaluating the asymptotic growth rate at the optimal strategy $\theta_t^\star = -\frac{1}{2}\Sigma^{-1}X_t$. From the completed-square expression, we have

$$\lambda^\star = \frac{1}{8}\int_E (\Sigma^{-1}x)^\top A(x)(\Sigma^{-1}x)\,dx.$$

Since $A(x) = cp(x)$, this becomes

$$\lambda^\star = \frac{1}{8}\int_E x^\top\Sigma^{-1}c\Sigma^{-1}x\,p(x)\,dx.$$

Using $X \sim N(0,\Sigma)$, we have $\mathbb{E}[X^\top\Sigma^{-1}c\Sigma^{-1}X] = \mathrm{Tr}(\Sigma^{-1}c\Sigma^{-1}\mathbb{E}[XX^\top]) = \mathrm{Tr}(\Sigma^{-1}c)$. Therefore, the robust-optimal asymptotic growth rate is given by

$$\boxed{\lambda^\star = \frac{1}{8}\,\mathrm{Tr}(c\Sigma^{-1}).}$$

PART II

Question 1

The general d-dimensional dynamics of an Ornstein-Uhlenbeck process (OU) are given by:

$$dX_t = B(m - X_t)dt + c^{\frac{1}{2}}dW_t$$

where the matrix $c^{\frac{1}{2}}$ is such that $c^{1/2}(c^{1/2})^\intercal = c$ where c is the instantaneous covariance matrix given in **Part I**. Note that we enforced in the previous part that $X_\infty \sim \mathcal{N}(0,\Sigma)$ for some given matrix Σ. Since $m \in \mathbb{R}^d$ is the long-run mean of the process, we also have $m = 0$. Therefore, the OU dynamics in our case reduce to

$$dX_t = -BX_t dt + c^{\frac{1}{2}}dW_t$$

In the minimax setup, "Nature" will only impact the drift term through the matrix B since the diffusion of the process depends on c which is fixed.

Question 2

For the d-dimensional Ornstein–Uhlenbeck process $dX_t = -BX_t\,dt + \sigma dW_t$, with $c = \sigma\sigma^\top$, the density $P(x,t)$ of X_t must satisfy the Fokker–Planck equation

$$\partial_t P(x,t) = \sum_{i=1}^{d}\sum_{j=1}^{d} B_{ij}\partial_{x_i}(x_j P) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d}\sigma\sigma^\intercal\partial_{x_i x_j}P$$

Since we require the invariant distribution to be $\mathcal{N}(0,\Sigma)$, the stationary density is the Gaussian density p. Stationarity means that $\partial_t P = 0$ and $P(x,t) = p(x)$. Therefore, after applying the product rule to the drift term, the stationary Fokker-Planck equation becomes $\mathrm{Tr}(B)p + (\nabla p)^\top Bx + \frac{1}{2}\,\mathrm{Tr}(c\nabla^2 p) = 0$.
Using the results $\nabla p(x) = -\Sigma^{-1}xp(x)$ and $\nabla^2 p(x) = (\Sigma^{-1}xx^\top\Sigma^{-1} - \Sigma^{-1})p(x)$, and after some tedious rearrangements, we obtain that this equation is satisfied if and only if the matrix B satisfies the Lyapunov condition:

$$\boxed{B\Sigma + \Sigma B^\top = c.}$$

Hence, Nature cannot choose an arbitrary drift matrix B: it must choose B subject to $B\Sigma + \Sigma B^\top = c$ in order to preserve the invariant distribution $N(0,\Sigma)$.

Question 3

We can show easily that the symmetry of the matrix c combined with a symmetric/skew-symmetric decomposition of the matrix ΣB gives us that the matrix B must be of the form $B = (\frac{1}{2}c + A)\Sigma^{-1}$ where A is any $d \times d$ skew-symmetric matrix. Any $d \times d$ skew-symmetric matrix has exactly $\frac{d(d-1)}{2}$ independent entries, hence so does B. Therefore, in the minimax game, we parametrize the matrix B as:

$$B_\beta = (\frac{1}{2}c + \beta_{\text{skew}})\Sigma^{-1}, \text{ where } \beta_{\text{skew}} = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$$

Question 4

In the implementation, the Agent is parameterized by an MLP $\varphi_\alpha : \mathbb{R}^2 \to \mathbb{R}$, with architecture $2 \to 64 \to 64 \to 64 \to 1$ and tanh activations (to ensure required smoothness). The trading strategy is then defined as the gradient of this MLP, namely $\theta_\alpha(x) = \nabla_x \phi_\alpha(x)$.
For Nature, since $d = 2$, the admissible perturbation of the OU drift has only one degree of freedom. We therefore parameterize a skew-symmetric matrix by one trainable scalar β, using the parametrization described above in question 3. This automatically enforces the Lyapunov constraint $B_\beta \Sigma + \Sigma B_\beta^\top = c$, so Nature only explores admissible OU dynamics preserving the invariant distribution $N(0, \Sigma)$. In practice, β is implemented as a bounded trainable parameter using a tanh transformation.
For fixed Agent and Nature parameters, the growth rate is estimated by Monte Carlo sampling $x^{(i)} \sim N(0, \Sigma)$ and computing

$$\widehat{g}(\alpha, \beta) = \frac{1}{M} \sum_{i=1}^{M} \left[-\theta_\alpha(x^{(i)})^\top B_\beta x^{(i)} - \frac{1}{2} \theta_\alpha(x^{(i)})^\top c \, \theta_\alpha(x^{(i)}) \right].$$

Note that this loss uses the ergodicity of the process X_t that we have discussed in Part 1. Note that another approach would have been to sampling paths from the dynamics of X_t and compute the discretized stochastic integrals along the paths. I chose to go with the second approach to stay in the "philosophy" of part1. The Agent tries to maximize this quantity, while Nature tries to minimize it. Hence the losses are $\mathcal{L}_{\text{Agent}} = -\widehat{g}(\alpha, \beta)$ and $\mathcal{L}_{\text{Nature}} = \widehat{g}(\alpha, \beta)$. Training is performed by alternating gradient steps: first updating Nature with the Agent frozen, then updating the Agent with Nature frozen.

Question 5

For the formulation of the losses, please see question 4. To handle the potential instabilities during training, we used a large number of epochs (3000), as well as large-samples for the Monte-Carlo estimator of $\widehat{g}(\alpha, \beta)$ in order to reduce the variance of the gradient steps ($M = 8192$). We also used small learning rate :

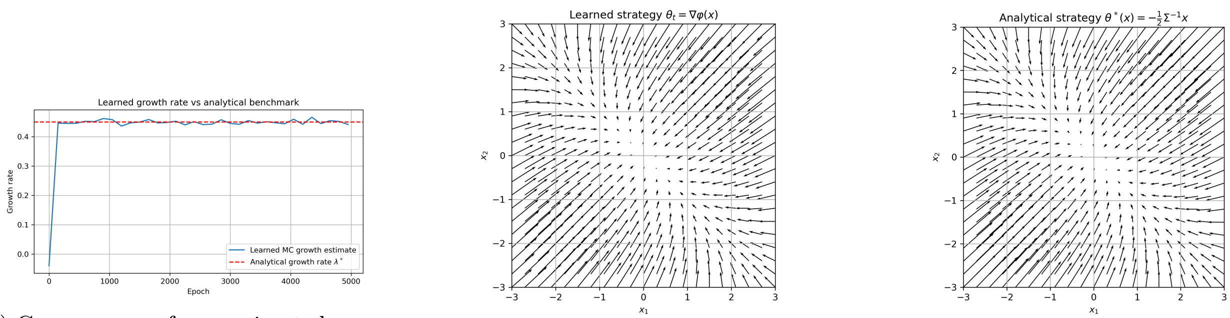
- $\eta_{\text{nature}} = 5 \cdot 10^{-4}$
- $\eta_{agent} = 10^{-3}$

Question 6

The first plot shows clear convergence of the growth rate approximated by the agent after a few hundred epochs to the true asymptotic growth rate $\lambda^\star$. The second and third plots show the vector fields of the trading strategies (respectively the Agent's approximation and the closed form solution $\theta_t(X_t) = -0.5\Sigma^{-1}X_t$). We see a strong similarity between both vector fields, where the largest differences can be observed in the bottom left and top-right corners of $[-3, 3]^2$.

Question 7

In the end, we evaluate the approximated growth-rate on a 20K-point sample of $\mathcal{N}_\infty$. We obtain $\widehat{\lambda^*} = 0.448799$ versus the closed form solution of $\lambda^* = 0.45$ (see our choice of c and Σ in the notebook leading to this value λ^*.) This gives a relative error of 0.2% which is arguably very reasonable. These results are obtained in approximately 1m30s of training on a local PC.



(a) Convergence of approximated asymptotic growth rate vs. epochs

(b) Vector field $\nabla \mathrm{MLP}(X_t)$

(c) Analytical vector field $\nabla\varphi(X_t)$

Figure 1: Some plots showing results